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LAN network

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Communication in local network

Local networks and wide area networks

Wide area network WAN :

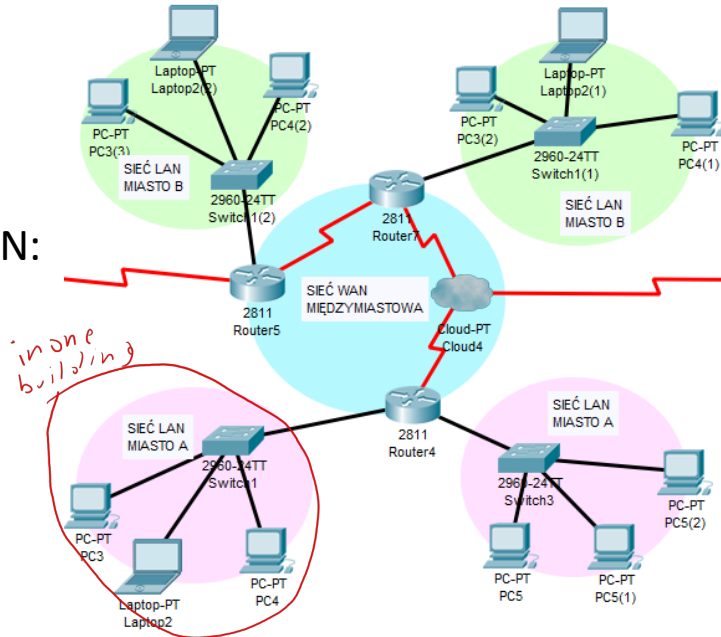
- It covers a larger area,

Metropolitan area network MAN:

- Area "city",

Local Area Network LAN :

- A small area,



SWITCHING...

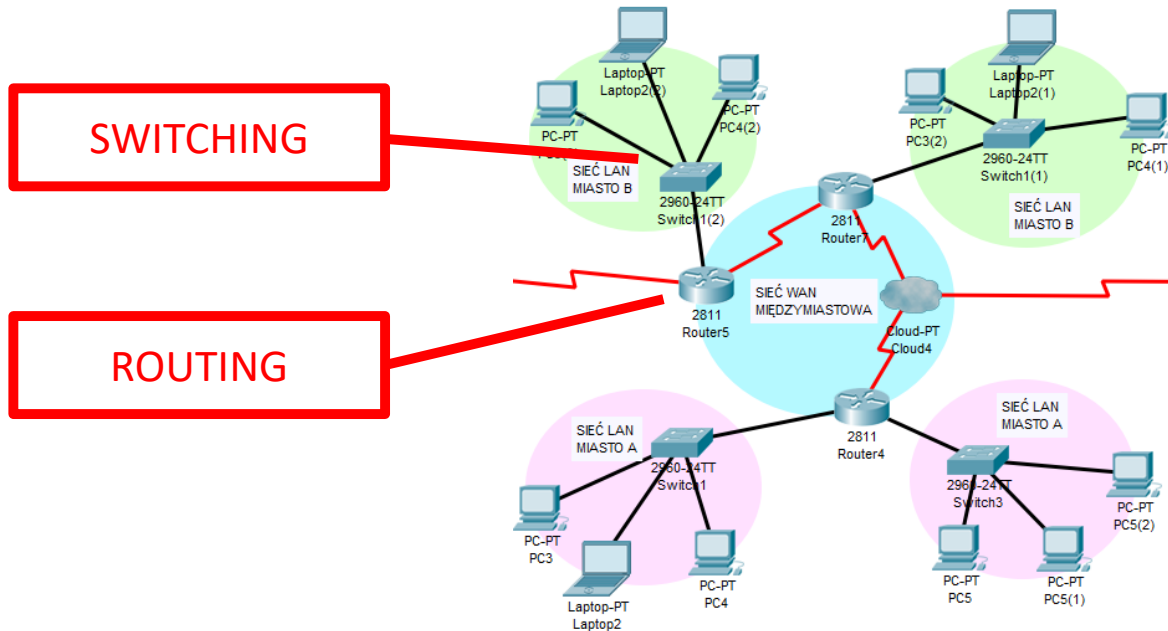
...is a method of forwarding frames on local networks

Routing (trasowanie)

Routing is...

... finding the best routes on the web.

Local networks and wide area networks



Switching vs routing

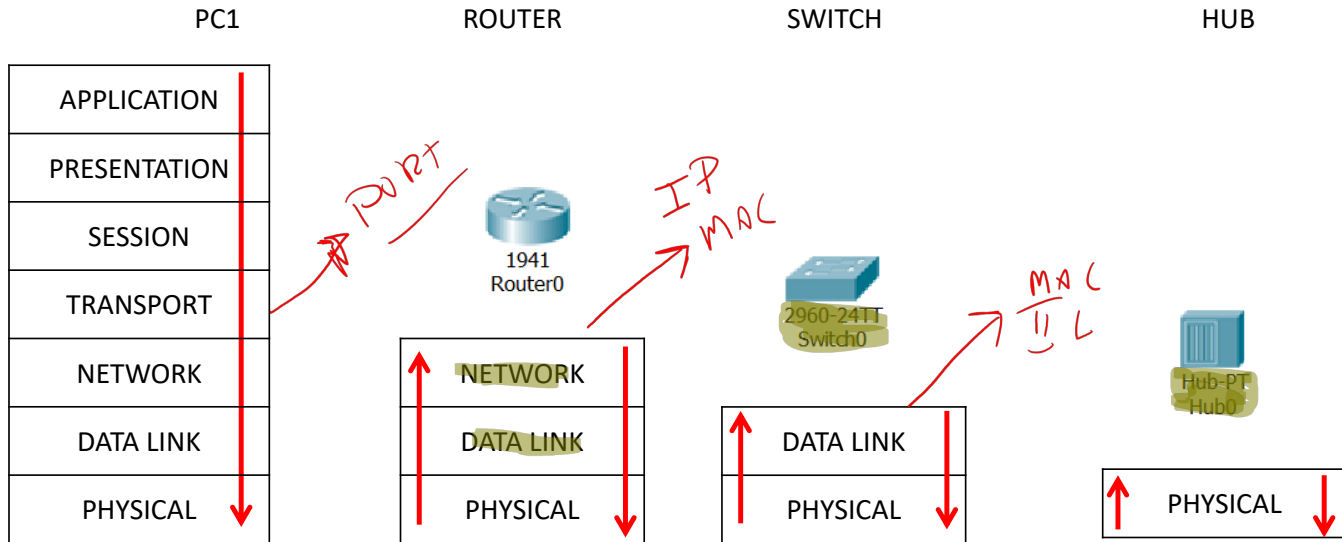
SWITCH / NETWORK SWITCH:

- It operates in layer II of the ISO / OSI model - the data link layer
- PDUs (protocol data units) are called "frames"

ROUTER:

- It operates in layer III of the ISO / OSI model - the network layer
- PDUs (Protocol Data Units) are called "packets"

Network devices and the model ISO/OSI



Network protocols

		MODEL OSI	PDU	Addres	Device	Protocols	MODEL TCP
Implementation	Software	7 Application				HTTP, DNS,	Application
		6 Presentation				SMTP, POP,	
		5 Session				SSL, SQL, DHCP	
	hardware	4 Transport	Segment TCP Datagram UDP	Port Number		TCP, UDP, SCTP	Transport
		3 Network	Packet	IP address	Router (Ruter)	IPv4, IPv6, ICMP	Internet
		2 Link layer	Frame	Physical address MAC	Switch (Przełącznik)	ARP, Ethernet, PPP, CDP	Network Access
		1 Physical			Hub (Koncentrator)	Bluetooth, DSL, 802.11	

TODAY

Topologies in local networks

Topologies in local networks - model of the connection system of various elements of a computer network.

Physical topology

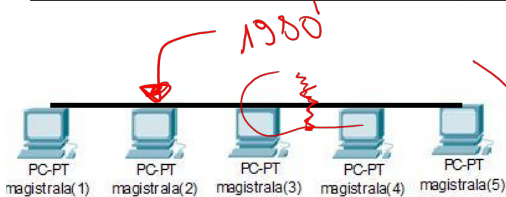
- describes the physical implementation of a computer network,

Logical topology

- describes how hosts communicate in physical topology,

Topologies in Ethernet

I. Connection with main line (bus)



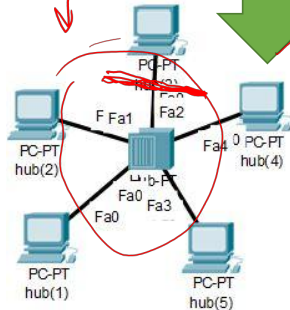
Topology:

- Physical: main line (bus)
- Logical: main line (bus)

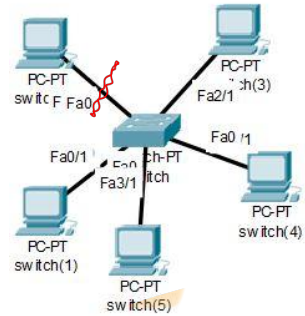
II. Adding a hub

Topology :

- Physical: star
- Logical: main-line



III. Adding a network switch



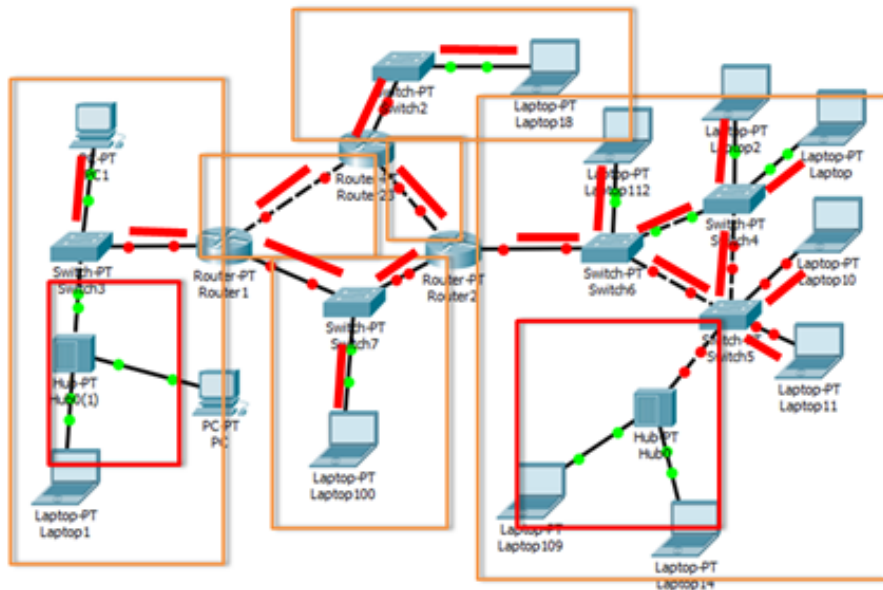
Topology :

- Physical: star
- Logical: pint-point

Broadcast domains vs collision domains

Broadcast domains
(domeny rozgłoszeniowe)

Collision domains
(domeny kolizji)



Ways of data transmission in networks

Unicast transmission (transmisja unicast):

- The frame is only sent to one device, 

Multicast transmission (transmisja multicastowa):

- The message is directed only to a selected group of hosts, 

Broadcast transmission (transmisja rozgłoszeniowa):

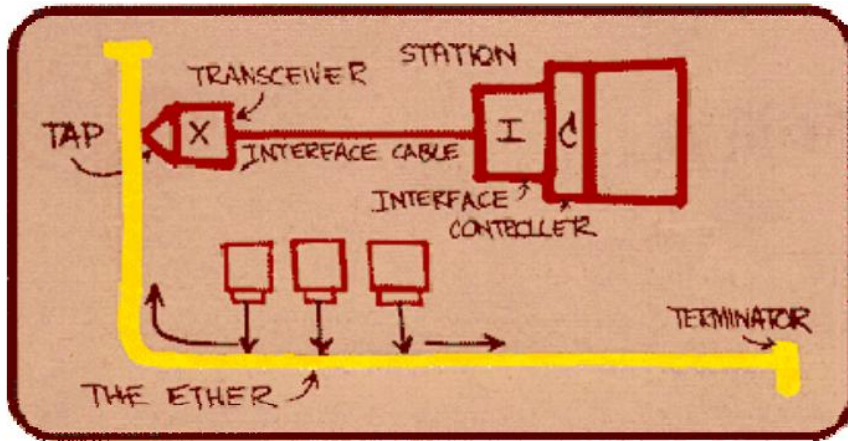
- A special address is used that allows all nodes to receive and process the frame, 

History of Ethernet protocol

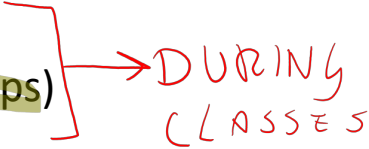
Network protocols

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		2 Link layer	Frame	Physical address MAC	Switch (Przełącznik)	Ethernet, ARP, PPP, CDP	Network Access
		1 Physical			Hub (Koncentrator)	Ethernet, DSL, 802.11	

The diagram drawn by Robert M. Metcalfe in 1976...



Ethernet - history

- 1973 - XEROX has created a prototype of a computer network.
 - 1976 – publication „Ethernet: Distributed Packet-Switching for Local Computer Networks”
 - 1980 – DIX (DEC, Intel, Xerox) "blue book" Ethernet specification
 - 1985 – IEEE 802.3 Ethernet standard
 - 1993 – Fast Ethernet Alliance (100 Mbps)
 - 1996 – Gigabit Ethernet Alliance (1000 Mbps)
 - 2000 – 10 Gigabit Ethernet Alliance
 - 2006 – Ethernet Alliance
- 

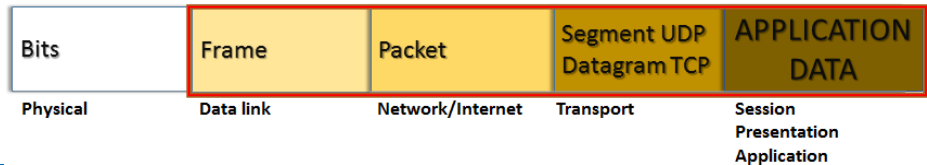
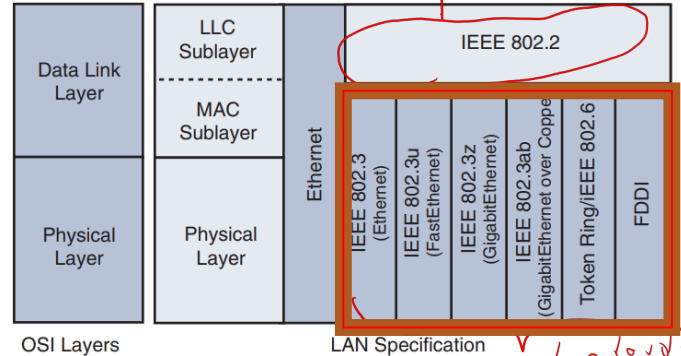
Source: www.intel.com

Ethernet in physical layer

Ethernet frame

There are two types of Ethernet frame:

- IEEE 802.3 (original version)
- Ethernet II DIX



Network mediums

- Copper wire



- Optical fiber



- Wireless media



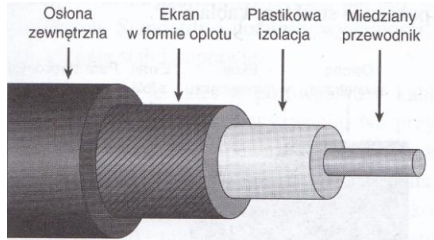
Ethernet standards

https://en.wikipedia.org/wiki/IEEE_802.3

Standard IEEE	Transfer	Medium type	Wiring	Max. Distance [m]
802.3a (thinnet)	10 Mbps	10BASE2	coaxial	>100m
802.3i	10 Mbps	10 BASE-T	UTP kat. 3/5	100
802.3u (Fast Ethernet)	100 Mbps	100 BASE-TX	UTP kat. 5 cable	100
		100 BASE-FX	multimode/ singlemode fiber	400 / 2 000
802.3z (Gigabit Ethernet)	1000 Mbps	1000 BASE-T	UTP kat. 5e cable	100
		1000 BASE-SX	multimode fiber	550
		1000 BASE-LX	singlemode fiber	2000
802.3ae (10Gigabit Ethernet)	10 Gbps	10G BASE-T	UTP kat. 6a/7 cable	100
		10G BASE-LX4	singlemode fiber	300
		10G BASE-LX4	Singlemode giber	10 000

Concentric copper cable

Concentric



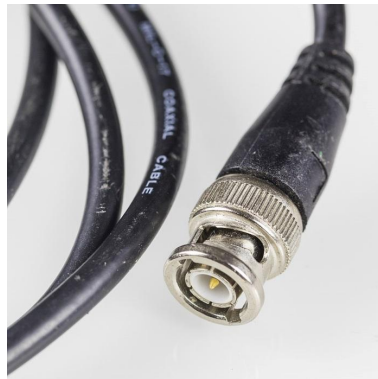
- Deprecated, standard 802.3a
- Ethernet in coax cable used to send for long distances (monitoring)



Coaxial cable - 10Base2

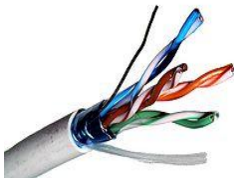
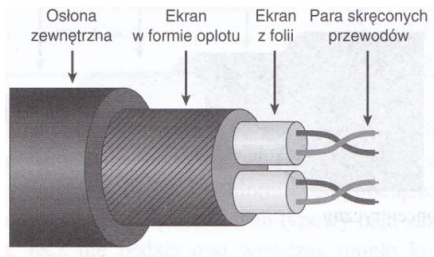
10Base2

- „cienki Ethernet” (ang. thin Ethernet, thinnet, thinwire)
- a network protocol related to the Ethernet family of standards operating in baseband at 10 Mbps,
- described in the 1985 IEEE 802.3a standard.



Twisted pair cable – categories

Twisted pair



	Used in	Up to	Frequency	Years
CAT 1	sound		1 MHz	1980s
CAT 2	Token Ring	4 Mbps	4 MHz	1980s
CAT 3	Ethernet	10 Mbps	10 MHz	1980s
CAT 5	FastEthernet	100 Mbps	100 MHz	1990s
CAT 5e		1000 Mbps	100 MHz	2000s
CAT 6	GigEthernet	10 Gbps	250 MHz	2000s
CAT 6a		10 Gbps	600 MHz	2000s
CAT 7		10 Gbps	600 MHz	2010s
CAT 8		> 25 Gbps	2 000 MHz	2010s

https://www.sopto.com.cn/sp_news/show-27560.html

Twisted pair cable – protection

UTP

Unshielded Twisted Pair

STP

Shielded twisted pair

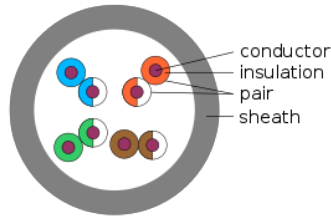
S/UTP

Shielded cable with unshielded twisted pairs

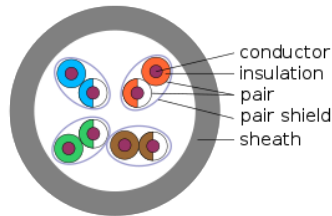
S/STP

Shielded cable with shielded twisted pairs

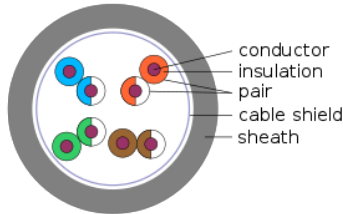
UTP



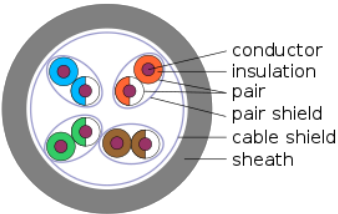
STP



S/UTP

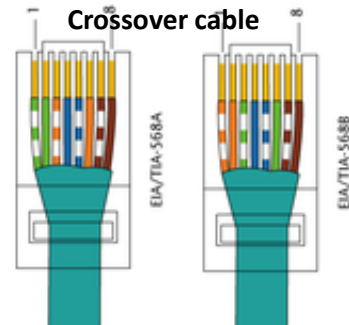
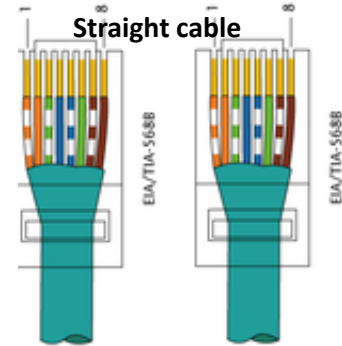
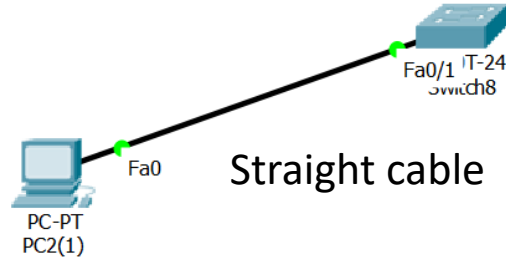
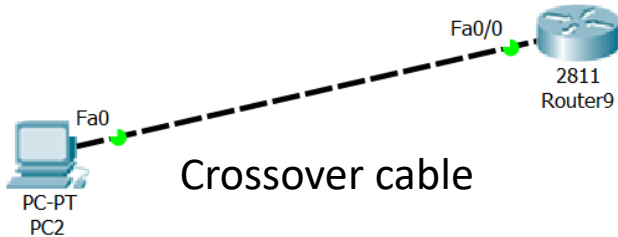


S/STP



<https://www.utepo.net/article/detail/utepo%20twisted%20b.html>

Twisted pair cable - Crossover and straight cable



	Hub	Switch	Router	Working station
Hub	crossover	crossover	straight	straight
Switch	crossover	crossover	straight	straight
Router	straight	straight	crossover	crossover
Working station	straight	straight	crossover	crossover

Twisted pair cable - AUTO MDI-X technology

AUTO MDI-X

- technology for the network device to automatically recognize whether the connected network cable is a crossover or a straight-through,
- thanks to this function, both a crossover and a straight-through cable can be connected to any 8P8C port of a network switch, network card or any other network device

Twisted pair cable – PoE

- Possibility of power supply via an ethernet connection
- A special switch is required!



Ethernet in data link layer

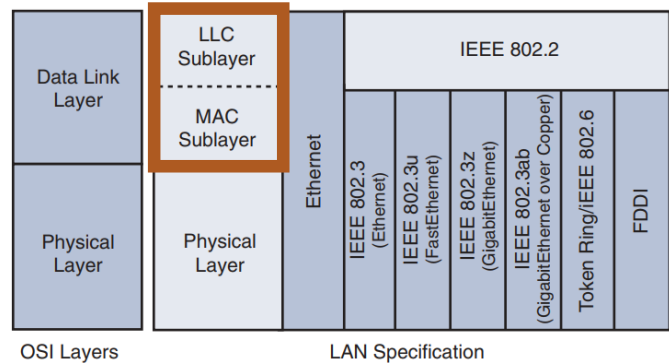
Ethernet in data link layer

LLC sublayer (802.2)

- Acts as an interface between the medium access control (MAC) sublayer and the network layer.
- Multiplexing mechanisms for several network protocols (e.g. IP, IPX)
- Rarely used in modern Ethernet

MAC sublayer (802.3):

- Handles framing, error detection (via CRC),
- Defines addressing (MAC addresses)
- Controls how devices access the physical medium (CSMA/CD)

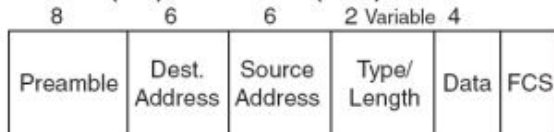


https://en.wikipedia.org/wiki/IEEE_802.3

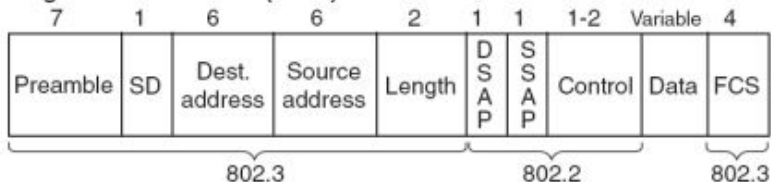
There are three types of Ethernet frame:

- Ethernet DIX = Ethernet II
- 802.2 Logical Link Control (LLC)
- Subnetwork Access Protocol (SNAP) extension

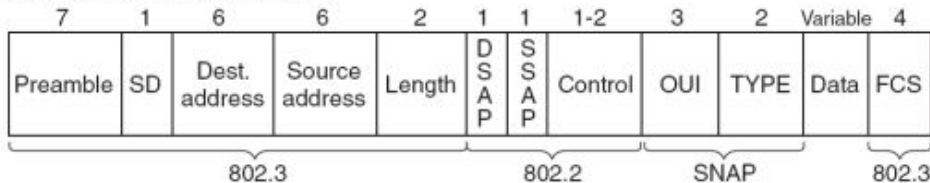
Ethernet (DIX) and Revised (1997) IEEE 802.3



Original IEEE Ethernet (802.3)

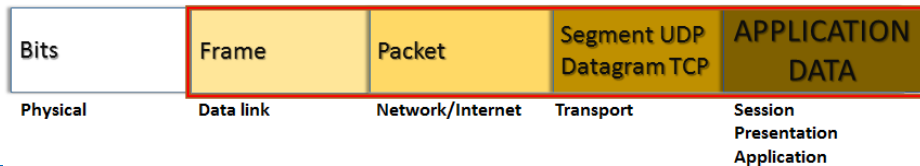


IEEE 802.3 with SNAP Header



Ethernet frame

Layer	Preamble	Start frame delimiter	MAC destination	MAC source	802.1Q tag (optional)	Ethertype (Ethernet II) or length (IEEE 802.3)	Payload	Frame check sequence (32-bit CRC)	Interpacket gap
	7 octets	1 octet	6 octets	6 octets	(4 octets)	2 octets	46-1500 octets	4 octets	12 octets
Layer 2 Ethernet frame	← 64–1522 octets →								
Layer 1 Ethernet packet & IPG	← 72–1530 octets →								← 12 octets →



Selected fields in an Ethernet frame

Preamble 56 bit combination (10101010...) allows devices in the network to easily synchronize receiver clocks, ensuring bit-level synchronization.

SFD start frame tag 10101011 that marks the end of a preamble and indicates the start of an Ethernet frame. The SFD is designed to interrupt the preamble bit pattern and signal the start of an actual frame

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Selected fields in an Ethernet frame

MAC addresses

- Each device in the ethernet network has its own unique address
- The address is 48bit

**producer
here:CISCO**

**Device
number**

00-60-2F-3A-07-BC

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Selected fields in an Ethernet frame

Frame addresses:

- **Unicast** specify a transmission linking one receiver and one sender.
Most of addresses in MAC pool

- **Multicast** use Ethernet frames with MAC addresses in the range
01:00:5e:00:00:00 - 01:00:5e:7f:ff:ff, e.g.:

- CDP/VTP- **0100.0ccc.cccc** multicast address

- STP - **0100.0ccc.cccd** BPDU multicast address

- **Broadcast** Layer 2 devices (switches) pass to all interfaces.

FFFF.FFFF.FFFF

Selected fields in an Ethernet frame

Field Ethertype or length

- 0-1500 - describes the maximum length of the datagram (Ethernet 802.3),
- >1536 - describes the EtherType value (Ethernet II DIX)
 - IPv4 0x0800
 - IPv6 0x86DD
 - ARP 0x0842
 - PROFINET 0x8892
 - EtherCAT 0x88A4
 - Wake-on-LAN 0x0842
 - VLAN 0x9100

HTTP
TCP
IPv4

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Selected fields in an Ethernet frame

Pole FCS - checksum in accordance with CRC-32

$$x^{32} + x^{26} + x^{23} + x^{22} + x^{16} + x^{12} + x^{11} + x^{10} + x^8 + x^7 + x^5 + x^4 + x^2 + x + 1$$

IPG (interpacket gap) - the minimum required interval between network frames

Layer	Preamble	Start frame delimiter	MAC destination	MAC source	802.1Q tag (optional)	Ethertype (Ethernet II) or length (IEEE 802.3)	Payload	Frame check sequence (32-bit CRC)	Interpacket gap
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How switch manages with
frames

Switch association table (MAC table)

- A table linking the switch port numbers to the MAC address.
- Basic switch operation control table.

Switch> show mac address-table

```
Switch>
Switch>show mac ?
      address-table  MAC forwarding table
Switch>show mac add
              Mac Address Table
-----
```

Vlan	Mac Address	Type	Ports
----	-----	-----	-----
1	0001.430a.76da	DYNAMIC	Fa6/1
1	0001.64a1.8537	DYNAMIC	Fa6/1
1	0001.9619.4ec0	DYNAMIC	Fa6/1
1	00d0.ba17.2dd6	DYNAMIC	Fa6/1
1	00d0.baed.b04a	DYNAMIC	Fa6/1

```
Switch>
```

ARP (Address Resolution Protocol)

The ARP protocol provides two basic functions:

- Mapping IPv4 addresses to MAC addresses
- Maintaining a mapping in ARP table

ARP table

- an array in RAM that connects network layer (IPv4) addresses with host hardware (MAC) addresses
- is stored in layer 3 devices (network)

Display the ARP table of the router and the computer

```
Router> show arp
Router> clear arp
```

```
Router>
Router>
Router>show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.20.2 - 00D0.FF6B.CE01 ARPA FastEthernet0
Router>
Router>
```

```
ubuntu@prod5:~$
ubuntu@prod5:~$
ubuntu@prod5:~$ arp -a
? (10.144.204.93) at 96:7d:5f:f5:f5:8d [ether] on ztbt06b7xf
? (10.144.3.167) at <incomplete> on ztbt06b7xf
? (10.144.41.245) at 96:d5:2f:ca:86:f7 [ether] on ztbt06b7xf
? (192.168.136.21) at <incomplete> on wlan0
_gateway (192.168.136.1) at 00:10:db:ff:10:00 [ether] on wlan0
ubuntu@prod5:~$
ubuntu@prod5:~$
ubuntu@prod5:~$
```

```
ip -s -s neigh flush all
```

```
Wiersz polecenia
C:\Users\user>arp /a

Interface: 192.168.1.101 --- 0x5
Internet Address Physical Address Type
192.168.1.1 94-0c-6d-aa-d8-50 dynamic
192.168.1.255 ff-ff-ff-ff-ff-ff static
224.0.0.2 01-00-5e-00-00-02 static
224.0.0.22 01-00-5e-00-00-16 static
224.0.0.252 01-00-5e-00-00-fc static
239.255.255.250 01-00-5e-7f-ff-fa static
255.255.255.255 ff-ff-ff-ff-ff-ff static

Interface: 192.168.246.1 --- 0xb
Internet Address Physical Address Type
192.168.246.254 00-50-56-fa-80-4e dynamic
192.168.246.255 ff-ff-ff-ff-ff-ff static
224.0.0.2 01-00-5e-00-00-02 static
224.0.0.22 01-00-5e-00-00-16 static
```

Cut-Through vs Store-and-Forward

Store-and-Forward Switching

- Uploading will wait for the entire frame to arrive before transmitting it
- The LAN switch will write every complete frame to the switch's memory buffers and check for errors before making a forwarding decision (CRC)
- If there are no errors, the frame will be sent to the destination address. Otherwise, the damaged frame will be rejected,

Cut-Through Switching

- The switch searches the first 6 bytes of the received frame (after the preamble),
- Based on the read MAC address, the switch will check the destination MAC address in its switch table, determine the port of the outgoing interface, and redirect the frame to the destination,
- No CRC error checking in the switching process,

Sources

1. Akademia sieci CISCO Exploration;
Semestr 1 Podstawy Sieci, Mark A. Dyne;
2. Akademia sieci CISCO Exploration;
Semestr 2 Protokoły i koncepcje routingu, Rick Graziani;
3. Akademia sieci CISCO Exploration; Semestr
3 Przełączanie sieci LAN i sieci bezprzewodowe, Wayne Lewis;
4. CCNA 200-125 Zostań administratorem sieci komputerowych,
Adam Józefiok;

